ANSWER KEY & MARKING SCHEME

CLASS 8 – CBSE CHEMISTRY QUESTION PAPER

Maximum Marks: 30

SECTION A: SHORT ANSWER QUESTIONS (8 Marks)

Q1. Chemical Reactions & Word Equations

- **Answer:**
  - *(a)* Magnesium + Oxygen → Magnesium Oxide
  - *(b)* Iron + Copper Sulphate → Iron Sulphate + Copper
- **Marking Scheme:**
  - **1 Mark:** Correct word equation for *(a)*.
  - **1 Mark:** Correct word equation for *(b)*.

Q2. Thermoplastics vs. Thermosetting Plastics

- **Answer:**
  - **Thermoplastic:** Linear or lightly cross-linked polymer chains; softens easily on heating and can be remolded repeatedly. *Example:* Polythene or PVC.
  - **Thermosetting Plastic:** Cross-linked polymer network; sets permanently when molded once and does not soften upon heating. *Example:* Bakelite or Melamine.
- **Marking Scheme:**
  - **1 Mark:** Structural/behavioral distinction between the two types.
  - **1 Mark:** Correct examples provided for both types (½ mark each).

Q3. Calorific Value of a Fuel

- **Answer:**
  - **Definition:** The amount of heat energy produced on complete combustion of 1 kg of a fuel is called its calorific value.
  - **SI Unit:** Kilojoule per kilogram (kJ/kg).
- **Marking Scheme:**
  - **1 Mark:** Correct definition specifying complete combustion of 1 kg fuel.
  - **1 Mark:** Correct SI unit (kJ/kg).

Q4. Displacement Reaction & Reactivity

- **Answer:**
  - **Definition:** A chemical reaction in which a more reactive metal displaces a less reactive metal from its salt solution.
  - **Observation & Reason:** No reaction occurs when copper is dipped in Zinc Sulphate solution because copper is less reactive than zinc and cannot displace zinc from its solution.

- **Marking Scheme:**
    - **1 Mark:** Accurate definition of a displacement reaction.
    - **1 Mark:** Stating "No reaction" along with the reactivity order reason ($Cu < Zn$).

SECTION B: MEDIUM ANSWER QUESTIONS & EQUATIONS (12 Marks)

Q5. Balancing & Completing Chemical Equations

- **Answer:**
    - *(a)* $2Na + 2H_2O \rightarrow \mathbf{2}NaOH + H_2 \uparrow$ *(Sodium Hydroxide)*
    - *(b)* $SO_2 + H_2O \rightarrow H_\mathbf{2}SO_\mathbf{3}$ *(Sulphurous Acid)*
    - *(c)* $Zn + H_2SO_4 \rightarrow ZnSO_\mathbf{4} + H_2 \uparrow$ *(Zinc Sulphate)*
- **Marking Scheme:**
    - **1 Mark:** Correct formula and balanced form for *(a)*.
    - **1 Mark:** Correct formula for product in *(b)*.
    - **1 Mark:** Correct formula for product in *(c)*.

Q6. Destructive Distillation of Coal

- **Answer:**
    - **Process:** Heating coal at high temperatures ($1000°C$ or higher) in the absence of air to break it down into various chemical components.
    - **Products & Uses (Any two):**
        1. **Coke:** Used as a reducing agent in metal extraction (e.g., steel manufacturing) and as a smokeless fuel.
        2. **Coal Tar:** Used to manufacture synthetic dyes, drugs, perfumes, plastics, and paints.
        3. **Coal Gas:** Used as an industrial fuel.
- **Marking Scheme:**
    - **1 Mark:** Explanation of destructive distillation (heating coal without air).
    - **2 Marks:** Two products with their correct industrial applications (1 mark for each product + use pair).

Q7. Metals & Non-Metals with Acids

- **Answer:**
    - *(a)* **Metals** generally react with dilute acids to produce metallic salts and evolve hydrogen gas. **Non-metals** generally do not react with dilute acids.
    - *(b)* **Gas:** Hydrogen gas (H_2).

Test: Bring a burning matchstick near the mouth of the test tube; hydrogen gas burns with a distinct **pop sound**.

- **Marking Scheme:**
    - **2 Marks:** Distinctive comparison of acid reactivity for metals vs. non-metals.

- **1 Mark:** Identifying Hydrogen gas and stating the "pop sound" test (½ mark each).

Q8. Candle Flame Zones & Temperature

- **Answer:**
  - **Diagram:**
    - *Outer Zone:* Blue color, complete combustion (Hottest zone).
    - *Middle Zone:* Yellow color, incomplete combustion (Luminous, moderately hot).
    - *Inner Zone:* Black color, unburnt wax vapors (Non-luminous, least hot).
  - **Hottest Zone & Reason:** The **outermost zone** is the hottest because it gets an abundant supply of oxygen from air, leading to complete combustion of fuel particles.
- **Marking Scheme:**
  - **1.5 Marks:** Neat, clearly labeled diagram highlighting all three zones.
  - **1.5 Marks:** Identifying the outer zone as hottest + explaining complete combustion due to oxygen availability.

SECTION C: LONG ANSWER & PRACTICAL QUESTIONS (10 Marks)

Q9. Corrosion & Chemical Equations

- **Answer:**
  - *(a)* **Rusting:** Slow degradation of iron metal when exposed to moist air over time forming hydrated iron oxide.

    Chemical Equation: $4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$
  - *(b)* **Greenish Coating on Copper:** Copper reacts slowly with atmospheric carbon dioxide, moisture, and oxygen to form a green coating of basic copper carbonate $\left(Cu(OH)_2 + CuCO_3\right)$.

    Chemical Equation: $2Cu + H_2O + CO_2 + O_2 \rightarrow Cu(OH)_2 + CuCO_3$
  - *(c)* **Prevention Methods (Any two):** Galvanization (zinc coating), painting/greasing, electroplating, or alloying.
- **Marking Scheme:**
  - **2 Marks:** Definition of rusting (1 mark) + balanced chemical formula/equation (1 mark).
  - **2 Marks:** Explanation of basic copper carbonate formation (1 mark) + chemical equation (1 mark).
  - **1 Mark:** Two valid prevention methods (½ mark each).

Q10. Polymers, Biodegradability & 5R Principle

- **Answer:**
  - *(a)* **Properties of Plastics (Any three):** Lightweight, non-reactive (corrosion resistant), strong and durable, poor conductor of heat and electricity, easy to mold.
  - *(b)* **Difference:**
    - **Biodegradable:** Substances that naturally decompose through biological processes carried out by microorganisms. *Example:* Paper, food waste, wood.

        - **Non-biodegradable:** Substances that cannot be broken down naturally by biological agents and persist in the environment for years. *Example:* Plastic bags, synthetic fibers, glass.
    - *(c)* **5R Principle: R**educe, **R**euse, **R**ecycle, **R**ecover, and **R**efuse.

- **Marking Scheme:**
    - **2 Marks:** Three distinct desirable properties of plastics (1.5 marks) + context of widespread use (0.5 mark).
    - **2 Marks:** Comparison between biodegradable and non-biodegradable along with suitable examples (1 mark each side).
    - **1 Mark:** Correctly listing all components of the 5R principle.